Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *CRISISLTLSUM* | ***Context:*** *Mexico City and Central America Bilingual Crisis Journalism*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The modality match is good: both the benchmark and target deployment operate on text-only tweets [Q2, Q18], and the elicitation explicitly marks this dimension as LOWER priority because no modality mismatch exists. However, the linguistic form of inputs is exclusively English [Q35, Q86], with preprocessing (lemmatization, lowercasing, fuzzy duplicate matching) implemented for English [Q129, Q136], and a 512-token BERT ceiling [Q92]. For a Spanish-language deployment, tokenization, lemmatization, and morphological handling must accommodate Spanish diacritics, accented-character variants, and informal diacritic-dropping in tweets — none of which is documented as supported. The 512-token sequence ceiling and tweet-text format are technically appropriate for the target. Per the elicitation, the linguistic-form mismatch is captured under IC and OF rather than IF; the modality dimension itself is well-aligned.

ID	Evidence
Q2	“We built CrisisLTLSum using a semi-automated cluster-then-refine approach to collect data from the public Twitter stream.” (p.1)
Q18	“CrisisLTLSum is developed through a two-step semi-automated process to create 1,000 local crisis timelines from the public Twitter stream.” (p.2)
Q35	“Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions.” (p.4)
Q86	“For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S.” (p.7)
Q92	“Although the BERT-based sequence classification model has a limitation when extracting long timelines due to its limited size of 512 positional embeddings, it performs better than the BERT-GRU sequence labeling model.” (p.7)
Q129	“To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemittized version of the keywords and the tweet's text.” (p.13)
Q136	“Here, the goal is to remove tweets from the same cluster with identical or similar text. The duplicate removal relies on a fuzzy string sequence-matching technique to compute the similarity between a pair of tweets. We simply go over each tweet sorted in chronological order and remove the ones that match any of the previous tweets in the same cluster with a matching score higher than dmatch.” (p.13)

Strengths

- Text-only tweet modality matches the target deployment exactly [Q2, Q18, elicitation IF-priority=LOWER].
- Standard tweet preprocessing pipeline (lemmatization, deduplication) [Q129, Q136] is methodologically applicable, even if its specific implementation is English-only.
- BERT-style 512-token sequence handling [Q86, Q92] is compatible with multilingual transformer backbones used for Spanish.

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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution match is acceptable at the modality level (text tweets) [Q2, Q18]. Within-modality differences (English vs. Spanish character set, diacritic handling, tweet-register orthography) require Spanish-specific preprocessing not documented in the benchmark [Q129].

IF-2: Yes — Mexican mobile/Twitter infrastructure supports the same text-tweet capture format. Mexico has ~152M mobile connections (1.2 phones/person) [WEB-14], indicating high source-tweet generation capacity. Central America country-level data was not retrieved [WEB-15].

IF-3: Domain-specific differences include: SASMEX-formatted seed tweets have a structured machine-readable form [WEB-7, WEB-8] that the benchmark's pipeline does not specifically handle; Spanish accented-character handling is required but undocumented; tweet-register diacritic-dropping is a known phenomenon requiring graceful handling.

IF-4: Form mismatches at the modality level are minor. The substantive linguistic-form mismatches (English-only preprocessing, English NER) materially affect IC and OF rather than IF per the elicitation framing.

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WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-8	@SASMEX X account provides structured earthquake seed tweets with no analog in benchmark content (https://x.com/sasmex)
WEB-14	Mexico mobile-phone penetration ~1.2 phones/person supports tweet-stream input form expectations (https://www.worlddata.info/america/mexico/telecommunication.php)
WEB-15	https://datahub.itu.int/data/?e=1&i=11624

Information Gaps

- Country-specific Central America mobile/internet penetration figures not retrieved
- Spanish tweet diacritic-dropping rates in crisis contexts not quantified